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Well-calibrated probabilities for prediction sets

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Abstract

Conformal prediction is a model-agnostic, distribution-free statistical framework that outputs prediction sets such that, for each prediction, the probability that the true label falls outside the set is less than a user-specified tolerance, ε. Notably, this guarantee holds conditionally on the calibration dataset and thus only prior to observing a new object. We present a method that assigns prediction probabilities to any set prediction. These probabilities approximate the true conditional probability that the label lies within the set, and converge to it with increasing data.

Venn-Abers predictors are (multi)-probabilistic predictors combining Venn-predictors and isotonic regression to output well-calibrated probabilities for binary prediction problems. We propose applying Venn-Abers to the constructed prediction sets from conformal prediction and, thereby, assigning a well-calibrated probability of the true label being in that set. We further introduce a novel self-consistent scoring rule for Venn–Abers predictors, which treats a predicted probability as being as likely as the event it postulates.

Additionally, we propose a set of strategies to reduce the computational requirements of the method while retaining all theoretical properties. We further investigate the trade-off between computation and performance with various calibration metrics. Lastly, we present one use case for the framework, introducing the concept of *actionable superclasses*, and demonstrate that the method can be used to support decision-making using a real-world high-stakes medical dataset where uncertainty quantification is essential.

Keywords: Inductive conformal prediction, Inductive Venn-Abers Predictor, Uncertainty Estimation, Uncertainty Quantification, Machine learning, High-stakes domains, Machine learning, Superclasses

1. Introduction

The increasing complexity of machine learning models, driven by deep architectures and vast training datasets, raises pressing questions about their trustworthiness. These concerns become particularly acute in high-stakes domains such as healthcare, finance, and criminal justice, where the reliability of model output can have profound ethical, legal, and societal implications. This underlines the need for reliable methods that estimate the uncertainty of predictions. Two frameworks within uncertainty quantification are conformal prediction and Venn-Abers predictors.

Conformal prediction constructs *valid* prediction sets for any *exchangeable* data distribution. A sequence of data points is exchangeable if it can be reordered without changing its joint distribution. Validity refers to the property that, prior to observing a new example, the probability that the true label of that example is contained in the prediction set is at least $1 - \varepsilon$, where ε is a pre-specified tolerance for error. $1 - \varepsilon$ is referred to as the *coverage probability*. The smoothed version of a conformal predictor is *exactly valid*, meaning that the coverage probability is exactly $1-\varepsilon$ Vovk et al. (2022). Crucially, the validity guarantees are applicable only before observing the features of each new example, a property commonly referred to as *marginal coverage* Angelopoulos and Bates (2021). After constructing a prediction set, but before observing the corresponding label, the probability that the label is contained in the set can be anywhere between 0 and 1. Naturally, for the empty set, the probability that it contains the true label is 0. Conversely, if the set spans the entire label space, the probability is 1. For large n, it follows that the coverage proportion for sets that are neither empty nor equal to the full label space must be approximately equal to:

$$1 - \varepsilon* = \frac{n(1-\varepsilon) - n_S}{n - n_S - n_\emptyset}, \tag{1}$$

where n is the number of examples, $n_\emptyset$ is the number of empty set predictions and n_S is the number of prediction sets equal to the full label space. Thus, for large n, the coverage guarantees provide information about the average coverage across all prediction sets, as well as across proper non-empty subsets of the label space.

While these guarantees are useful, we would like to be able to state how confident we are that a constructed prediction set contains the true label. Venn–Abers predictors provide well-calibrated prediction probabilities. A probabilistic predictor is said to produce well-calibrated predictions if it approximates *perfect calibration*, meaning that the expected value of an outcome is almost surely equal to the predicted probability Vovk and Petej (2012). Importantly, the expected value is conditional on a category defined by the predictor, i.e. the equivalence class into which the predictor places the example. The original Venn-Abers framework is only applicable to binary classification problems. However, we assert that it can be applied to any classification problem by considering binary splits of the label space. The prediction sets produced by conformal prediction implicitly define such splits; each element of the label space will either be included in or excluded from the prediction set. For each possible binary split of the label space, a Venn-Abers predictor is calibrated on the original features and a new label which takes the value 1 if the original label belongs to the prediction set and 0 otherwise. The output of this Venn-Abers is a well-calibrated prediction probability of the label being in the set.

The rest of this paper is organized as follows. Section 2 outlines the prerequisites for the work. Section 3 introduces the method, a new scoring rule and three strategies for reducing the computational requirements. Section 3 also presents an approach for utilizing the prediction probabilities on a real-world dataset. Section 4 reviews related work that situates our contribution within the broader literature. Section 5 reports and analyzes the results. Sections 6 and 7 provide a discussion of the findings and directions for future work. Finally, Section 8 concludes the paper.

2. Prerequisites

2.1. Inductive conformal predictors

A conformal predictor (CP) is a set predictor that is constructed using a nonconformity measure[1]. Intuitively, this is a function that quantifies how unusual an example is, with respect to previously observed examples. To describe CPs formally, two measurable spaces, $\boldsymbol{X}$ and $\boldsymbol{Y}$, are required, referred to as the object space and the label space, respectively. Their cartesian product $\boldsymbol{Z} := \boldsymbol{X} \times \boldsymbol{Y}$ is called the example space. We often abbreviate examples as $z := (x, y) \in \boldsymbol{Z}$. As this work is only concerned with what is known as inductive CPs (ICPs), we define these formally, and refer to Vovk et al. (2022) regarding full, or transductive CPs.

An inductive nonconformity measure is a function $A : \boldsymbol{Z}^* \times \boldsymbol{Z} \to \overline{\mathbb{R}}$ that, given a sequence of training objects $z_1, \ldots, z_m$, assigns a nonconformity score to a test object x and a postulated label y as $\alpha^y(x) := A((z_1, \ldots, z_m), (x, y))$. This measures how strange or unusual the example (x, y) is with respect to $z_1, \ldots, z_m$.

Given a training set $z_1, \ldots, z_l$, we construct an ICP as follows.

1. Divide the training set of size l into a proper training set of size m and a calibration set of size $k = l - m$.

2. Define an inductive nonconformity measure using $z_1, \ldots, z_m$. Common choices include training a machine learning model μ and applying $A((z_1, \ldots, z_m), (x, y)) = |\hat{y} - y|$, where $\hat{y} = \mu(x)$, and y is the label associated with x.

3. For a test object x, output the prediction set

$$\Gamma^\varepsilon(z_1, \ldots, z_l, x) := \left\{ y \in \boldsymbol{Y} : \frac{|\{j = m+1, \ldots, l : \alpha_j \geq \alpha^y\}| + 1}{l - m + 1} > \varepsilon \right\}, \tag{2}$$

where the nonconformity scores are computed as

$$\begin{aligned} \alpha_j &:= A((z_1, \ldots, z_m), z_j) \\ \alpha^y &:= A((z_1, \ldots, z_m), (x, y)), \end{aligned}$$

where ε is the significance level, i.e. the tolerance for error. When the training set is clear from context, we often write $\Gamma^\varepsilon(x)$ for the prediction set in 2.

1. It is also possible, and in some cases preferable, to use a conformity measure, whose purpose is the opposite; to quantify how typical an example is.

The fraction in the set construction 2 is called the p-value for (x, y). Constructing $\Gamma^{\varepsilon}(x)$ may be seen as testing the hypothesis that x has the label y for each possible label in the label space, and including those labels that cannot be discarded at significance level ε.

It is common to treat the borderline cases $\alpha_j = \alpha^y$ separately by introducing auxiliary randomness in the form of a random number τ, which is independent and distributed uniformly on $[0, 1]$. The difference is that the prediction set includes those labels for which the smoothed p-value

$$\frac{|\{j = m+1, \ldots, l : \alpha_j > \alpha^y\}| + \tau(|\{j = m+1, \ldots, l : \alpha_j = \alpha^y\}| + 1)}{l - m + 1} \tag{3}$$

exceeds the significance level ε Vovk et al. (2022).

It is convenient to state the training-conditional validity guarantee of ICPs in terms of the function

$$\Gamma(z_1, \ldots, z_l) = \{(x, y) \in \boldsymbol{Z} : y \in \Gamma(z_1, \ldots, z_l, x)\}, \tag{4}$$

where Γ is a set predictor and $z_1, \ldots, z_l$ is the training set. Assuming that examples are generated independently from the same distribution Q, we say that Γ is (ε, δ)-valid if, for any probability distribution Q on $\boldsymbol{Z}$,

$$Q^l\{(z_1, \ldots, z_l : Q(\Gamma(z_1, \ldots, z_l)) \geq 1 - \varepsilon\} \geq 1 - \delta. \tag{5}$$

In other words, (ε, δ)-validity means that with probability at least $1 - \delta$, the coverage probability is at least $1 - \varepsilon$. It is shown (as a consequence of Corollary 4.9) in Vovk et al. (2022) that any ICP Γ whose calibration set has size $h := l - m$, can be turned into an (ε, δ)-valid set predictor by adjusting the significance level. Specifically, the set predictor

$$\Gamma^{\varepsilon - \sqrt{\frac{\ln(1/\delta)}{2h}}} \tag{6}$$

is (ε, δ)-valid for any $\varepsilon \in (\sqrt{\frac{\ln(1/\delta)}{2h}}, 1)$. Note that, since the significance level must be positive, this implies a lower bound on the calibration set size

$$h > \frac{\ln(1/\delta)}{2\varepsilon^2}. \tag{7}$$

For $\varepsilon = \delta = 0.05$, we need at least 600 calibration examples, while $\varepsilon = \delta = 0.01$ requires at least 23026 calibration examples. In summary, for a sufficiently large training set split into a proper training set and a calibration set of size h, the adjusted ICP predictor includes the true label with probability at least $1 - \varepsilon$, except with probability at most δ Vovk (2012).

2.2. Venn Predictors and Inductive Venn-Abers Predictors (IVAPs)

A multiprobabilistic predictor is a predictor whose output consists of multiple probability distributions on the label space $\boldsymbol{Y}$. We briefly discuss a class of multiprobabilistic predictors called Venn predictors. Conceptually, for a test object $x \in \boldsymbol{X}$, they work by dividing the training objects into categories by some criterion, and then using the frequency of labels in the chosen category (formally a Venn Taxonomy) of x as probabilities for the label of the test object. We omit the formal definition of Venn taxonomies and predictors and refer to Vovk et al. (2022).

The key concept for our purposes is the validity of multiprobabilistic predictors. We say that a random variable P taking values in $\boldsymbol{P}(\boldsymbol{Y})$ (the set of probability distributions on $\boldsymbol{Y}$) is perfectly calibrated for a random variable Y taking values in $\boldsymbol{Y}$ if, for each $y \in \boldsymbol{Y}$,

$$\mathbb{P}(Y = y \mid P) = P\{y\} \quad \text{a.s.} \tag{8}$$

The output of a Venn predictor is the multiprobabilistic prediction $P := (P_y : y \in \boldsymbol{Y})$, where $P_y \in \boldsymbol{P}(\boldsymbol{Y})$, and the key property of Venn predictors is that there exists a selector S (a $\boldsymbol{Y}$-valued random variable) such that P_S is perfectly calibrated for y under any power distribution (Proposition 6.1 in Vovk et al. (2022)). Intuitively, this means that at least one of the, at most $|\boldsymbol{Y}|$, probability distributions is perfectly calibrated. Which one is unknown, but in practice, if the number of old examples in each category is large, these distributions tend to be close to one another, so we can expect them or some aggregation of them to be well-calibrated (i.e. they satisfy 8 approximately).

IVAPs are Venn predictors for binary predictions that use isotonic regression to create taxonomies. Isotonic regression is a non-parametric technique for fitting data under a monotonicity constraint, typically non-decreasing Dykstra (1983); Barlow et al. (1972), and is efficiently implemented via the pair-adjacent violators algorithm (PAVA) Ayer et al. (1955).

To construct an IVAP:

1. Split the training set of size l into a proper training set of size m and a calibration set of size $k = l - m$.

2. Train a binary scoring classifier on the proper training set to obtain a scoring function $S : X \to \mathbb{R}$.

3. Compute scores $s_1, \ldots, s_k$, where $s_i := S(x_i)$ for calibration objects $x_1, \ldots, x_k$.

4. For a test object x, compute its score $s := S(x)$, then:

   - Fit isotonic regression to $(s_1, y_1), \ldots, (s_k, y_k), (s, 0)$ to obtain the calibrator f^0.
   - Fit isotonic regression to $(s_1, y_1), \ldots, (s_k, y_k), (s, 1)$ to obtain the calibrator f^1.

5. Construct the multiprobabilistic prediction for the label y of x is $(p_0, p_1) := (f^0(s), f^1(s))$.

Either f^0 or f^1 satisfies 8, which one depends on whether the true label is 0 or 1. For large datasets, p_0 and p_1 converge and both satisfy 8, at least approximately. By construction, $p_0 < p_1$. For practical use, one may combine them into a single probabilistic prediction. A common choice, minimizing expected log loss Vovk et al. (2022), is:

$$p := \frac{p_1}{1 - p_0 + p_1} \tag{9}$$

In section 3.2 we introduce an alternative scoring rule.

3. Method

This section begins with a formal description of the proposed method, which combines the CP and Venn-Abers frameworks described in Section 2. It then introduces the motivation for the self-consistent Venn scoring rule and provides its mathematical derivation. Subsequently, three strategies for mitigating computational requirements are presented. Next, we introduce the concept of actionable superclasses to demonstrate a use case of the method. Lastly, the experimental setup is described. While the formal description references CP and Venn-Abers generally, the remainder of the paper references ICP and IVAP specifically, since these variants are used in all implementations. An algorithmic description of the method can be found in Appendix A.2 for readers interested in implementation details.

3.1. Formal description

Let $\mathbf{Z} = \mathbf{X} \times \mathbf{Y}$ be an example space, where $\mathbf{X}$ is an object space and $\mathbf{Y}$ is a label space.

Examples are denoted by $(x_i, y_i) \in \mathbf{Z}$ and the label space is assumed to be a non-empty finite set $A = \{a_1, \ldots, a_k\}$, so that $\mathbf{Y} = A$.

Partition the power set $\mathcal{P}(A)$ into 2^{k-1} pairs of subsets, where each pair consists of a subset of A that does not contain a_1, and its complement with respect to A:

$$\Pi_A = \Big\{\{B, B^C\} : B \subseteq A,\ a_1 \notin B\Big\} \tag{10}$$

Index the pairs in Π_A by defining:

$$\text{index}_{\Pi_A}(\{B, B^C\}) = 1 + \sum_{i=1}^{k} 2^{k-i}\mathbf{1}_B(a_i) \tag{11}$$

where $\mathbf{1}_B(a_i)$ is an indicator function taking the value 1 if a_i is in B and 0 otherwise. Let the index assigned to each pair be inherited by both of its elements, resulting in two sequences $(B_j)_{j=1}^{2^{k-1}}$ and $(B_j^C)_{j=1}^{2^{k-1}}$.

Given a sequence of examples $(x_i, y_i)_{i=1}^{n}$, define binary outcomes y_i^j for each $i \in \{1 \ldots, n\}$ and $j \in \{1, \ldots, 2^{k-1}\}$ by letting:

$$y_i^j = \begin{cases} 0 & \text{if } y_i \in B_j^C, \\ 1 & \text{if } y_i \in B_j \end{cases} \tag{12}$$

Let C_{VA} and C_{CP} denote two disjoint calibration sets, used respectively for the Venn–Abers predictors and for conformal prediction. Associate a Venn–Abers predictor with each partition $\{B, B^C\} \in \Pi_A$ using C_{VA} across all test examples $(x_i)_{i=1}^{n-m}$, yielding $n \times 2^{k-1}$ pairs of calibrated probabilities $(p_{0,i}^j, p_{1,i}^j)$.

Apply CP to the sequence $(x_i, y_i)_{i=1}^{n-m}$ using the calibration set C_{CP}, setting the tolerance for error to ε. To each prediction set $\Gamma^\varepsilon(x_i)$, assign the calibrated probabilities $(p_{0,i}^j, p_{1,i}^j)$ if $\Gamma^\varepsilon(x_i) = B_j$ and $(1 - p_{1,i}^j, 1 - p_{0,i}^j)$ if $\Gamma^\varepsilon(x_i) = B_j^C$ for some $j \in \{1, \ldots, 2^{k-1}\}$.

This sequence of operations yields, for each pair of subsets in Π_A, a corresponding Venn-Abers predictor

$$\mathrm{V^j}(x_i) = \big(f^{0,j}(s^j(x_i)),\ f^{1,j}(s^j(x_i))\big), \tag{13}$$

where $f^{0,j}$ and $f^{1,j}$ are isotonic calibrators and s is the score of the binary base classifier. Each predictor is associated with an oracle selector O^j that is perfectly calibrated. Specifically, for a prediction set $\Gamma^\varepsilon(x_i)$, O^j satisfies

$$O^j(V^j(x_i)) = \mathbb{P}(y_i \in \Gamma^\varepsilon(x_i) \mid O^j(V^j(x_i)). \tag{14}$$

In large samples, both $f^{0,j}$ and $f^{1,j}$ are well-calibrated and either of them could replace O^j and still satisfy 14 approximately.

3.2. Self-Consistent Venn Scoring Rule

We are given a Venn predictor producing, for an object x_{n+1}, a pair of probability outputs (p_0, p_1). Here p_0 denotes the probability assigned to the event $y_{n+1} = 1$ under the hypothetical label $y_{n+1} = 0$, and p_1 denotes the corresponding probability under the hypothetical label $y_{n+1} = 1$. Any single-valued probability p chosen from this interval must satisfy

$$p \in [p_0, p_1]. \tag{15}$$

A convenient way to enforce this constraint is to express p as a convex combination of the endpoints:

$$p = p_0 + (p_1 - p_0)\, w, \qquad w \in [0, 1]. \tag{16}$$

We now interpret the mixing weight w not as an arbitrary parameter but as the expected value of an underlying Bernoulli indicator. Specifically, let

$$W = \begin{cases} 1, & \text{if } y_{n+1} = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{17}$$

A natural requirement is that the mixing weight w reflects the expected value of the Bernoulli variable W across the two Venn scenarios. Consequently,

$$w = \mathbb{E}[W] = \Pr(W = 1). \tag{18}$$

Because the value $\Pr(W = 1)$ is precisely the true probability $\Pr(Y = 1)$, and because a calibrated predictor must satisfy $p = \Pr(Y = 1)$, we obtain the self-consistency condition

$$w = p. \tag{19}$$

Substituting $w = p$ into the convex combination equation yields the fixed-point equation

$$p = p_0 + (p_1 - p_0)\, p. \tag{20}$$

Solving for p gives

$$\boxed{p = \frac{p_0}{1 - p_1 + p_0}.} \tag{21}$$

This expression defines a unique fixed point in the interval $[p_0, p_1]$ and can be interpreted as the self-consistent expected probability of the event $Y = 1$. Because it weighs the prediction probabilities according to the probability of their respective postulates, it should reflect the expected value of Y and therefore minimize the expected calibration error. We refer to this scoring rule as the *self-consistent scoring rule*. In the remaining part of the paper when we reference a single IVAP prediction probability, this should be interpreted as p in equation (21).

3.3. Strategies for reducing computational overhead

Training one IVAP model for each prediction set introduces the potential for significant computational requirements. The full implementation generates 2^{k-1} prediction problems and consequently necessitates training an equal number of IVAP models. In most experimental settings, it is enough to train models only for the sets being produced by the ICP on the test set, since no other set predictions need to be evaluated. In such settings, the worst-case scenario requires $min(2^{k-1}, n_{test})$ prediction problems. Indeed, for large k, n and models, this may deteriorate the entire framework. To circumvent this problem, we suggest several computationally efficient methods and demonstrate that it is possible to reduce training time substantially. The first two methods are designed specifically for deep learning frameworks where computational requirements are most demanding. The presented methods rely on the assumption that the trained ICP has already learned useful information about the new binary classification tasks.

All algorithms trained through gradient descent are, in theory, capable of resuming training from a previous state through, e.g., weight initialization. In deep learning, this is common practice, where, e.g., large pre-trained models are frequently used as backbones of custom models Goodfellow et al. (2016). Thus, provided that the ICP model was trained using gradient descent and the IVAP models use the same underlying model, we propose swapping the multiclass classification layer of the base ICP model for a binary classification layer and initialize all other model parameters with the base ICP model parameters. We refer to this strategy as the *fine-tune* strategy. Alternatively, we propose freezing the base ICP model and training only a new binary classification layer to speed up each training epoch. We refer to this strategy as the *layer freeze* strategy.

Lastly, we suggest a strategy that avoids retraining altogether. Here, the heuristic probability estimate for the label being inside the set is simply the aggregated heuristic prediction probabilities across the labels inside the set assigned by the base ICP model. These are calibrated with an IVAP. We refer to this strategy as the *probability pooling* strategy. Note that the heuristic uncertainty estimates (i.e. softmax scores) of the labels included in the set have passed the threshold of the ICP, meaning that the aggregated score for the test point is at least $|\Gamma(x)| * q^*$, where $|\Gamma(x)|$ is the size of the prediction set and q^* is $(1-\varepsilon)$-quantile of the calibration. Further, the set contains the classes with the highest prediction probabilities of the base model which means their sum will also be larger than $\frac{|\Gamma(x)|}{k}$, where k is the size of the full label space. Consequently, there is a lower bound on the heuristic uncertainty estimations of $y = 1$ that will be the input to the IVAPs in the probability pooling strategy,

$$p(y = 1 \mid x)_{heuristic} > max(|\Gamma(x)| * q^*, \frac{|\Gamma(x)|}{k}). \tag{22}$$

Importantly, there is no such limit on the calibration points. The potential consequences of this observation are described in the discussion section.

3.4. Actionable Superclasses

The framework presented in this article is broadly applicable across a variety of real-world scenarios. In order to demonstrate the practical utility of the framework, we introduce the

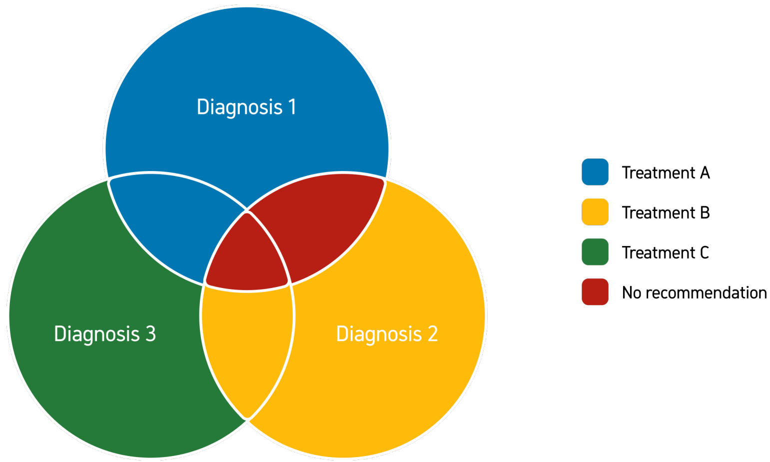

Figure 1: Example of relationship between superclasses (treatments) and labels (diagnosis)

notion of *actionable superclasses*. In the real-world, any class of objects is a member of multiple superclasses. Across all of these classes, their respective superclasses necessarily exhibit partial overlap. For example, a dog is both a pet and a four-legged animal, a lion is not a pet but a four-legged animal and parrot is a pet but not a four-legged animal. However, for most decisions, the majority of superclasses are not relevant. We therefore introduce the notion of *actionable superclasses*. Figure 1 exemplifies a simple label space with three labels (diagnosis) and three superclasses (treatments).

Treatment A can be used to treat both diagnoses 1 and 3. Therefore, a prediction set {diagnosis 1, diagnosis 3} with sufficiently high probability can provide clinically relevant decision support for treatment A. We refer to this notion as *actionable prediction sets*. Note that it is not possible to use the superclasses themselves as labels for classification in a standard multiclass setting, because individual labels may belong to multiple superclasses. Formally, this occurs when the hierarchy of classes and superclasses is represented by a directed acyclic graph (DAG) that does not form a tree structure. In order to utilize this conceptualization, experts need to define actionable superclasses of diagnosis. In 3.5, we introduce a real-world dataset where such an effort has been made. In many clinical settings, this conceptualization is a simplification of the decision making process. For decision-making processes that incorporate additional considerations, such as cost weighting and expected clinical outcomes that vary with the treatment, we propose that this concept can be relevant while further methodological advancements will be necessary.

3.5. Experimental set-up

Since we assert that the proposed framework yields well-calibrated probabilities as the size of the calibration set increases, we evaluate this claim by training random forest models on a large simulated dataset of size $N = 512{,}000$. Let $X \sim \text{Bern}\left(\frac{1}{2}\right)$. The conditional distribution of Y given X is

$$P(Y \mid X=0) = \left(\tfrac{37}{60}, \tfrac{1}{3}, \tfrac{1}{20}\right), \quad P(Y \mid X=1) = \left(\tfrac{1}{20}, \tfrac{1}{3}, \tfrac{37}{60}\right).$$

Further, we evaluate the method on two standard image-classification benchmarks: MNIST and CIFAR-10. MNIST consists of 70,000 grayscale images of handwritten digits (28×28) LeCun et al. (1998), while CIFAR-10 contains 60,000 color images (32×32) across ten object classes Krizhevsky et al. (2009). In both cases, our aim is not to achieve state-of-the-art accuracy but to assess how well the proposed framework integrates with deep learning models of varying complexity and to compare calibration performance across the presented strategies. Each dataset was evaluated with metrics averaged over four randomized splits. Each split was divided into 5 sets: training, validation, cp calibration, ivap calibration & test. The number of datapoints in the splits was (35000/5000/3750,11250,5000) for MNIST and (25000/5000/3750,11250,5000) for CIFAR). For both datasets and for both CP and IVAP, the models were convolutional neural networks trained with stochastic gradient descent, using the following hyperparameters: 30 epochs, a learning rate of 0.01, and a batch size of 256 for MNIST; and 100 epochs, a learning rate of 0.005 with momentum 0.9, and a batch size of 64 for CIFAR-10. We applied the four strategies described in the Method section—full retraining, fine-tuning, layer freezing, and probability pooling—to enable a consistent comparison of calibration behavior. For MNIST, three values of ε were applied, $0.015, 0.01$ and 0.05. For CIFAR-10, $\varepsilon = 0.2$. All training parameters and data-split seeds are available in the GitHub repository referenced in Section 8 to ensure full reproducibility.

Lastly, we demonstrate the method on real-world clinical data from a high-stakes domain. Cortes-Gomez et al. (2025a) (pre-print) applied conformal prediction on the clean Fitzpatrick17k Skin Tone dataset Groh et al. (2021), which provides overlapping actionable superclasses from 48 labels in a high-stakes domain. This is a subset of the full Fitzpatrick17k dataset Groh (2021) and contains the most common skin tone types. In Cortes-Gomez et al. (2025a), just as in the toy example in figure 1, the superclasses have been created by experts for diagnoses with shared downstream actions. This means that if all labels in a set prediction belong to a certain superclass, decision support systems could provide concrete recommendations despite uncertainty in the exact condition. As we do not have access to the models, we applied the probability pooling strategy on the softmax scores provided on the associated GitHub by Cortes-Gomez et al. (2025b).

4. Related Work

To the best of our knowledge, no other method assigns well-calibrated prediction probabilities to prediction sets. van der Laan and Alaa (2024) proposed a method that combines Venn-Abers predictors, extended for regression, with conformal prediction. This method generates calibrated point predictions and intervals with prediction-conditional validity. In this framework, a prediction $f(x)$ functions as a scalar dimension reduction of an object x and thus, prediction-conditional validity is a relaxation of the infeasible objective of object-conditional validity. Here, the prediction refers to the calibrated output of the Venn-Abers predictor. The two frameworks are similar in that they import the properties of Venn-Abers to the conformal prediction framework to provide conditional guarantees. However, their work provides a validity guarantee rather than assigning probabilities to the output of a conformal predictor. As discussed in the Future Work section, we plan to extend our framework to produce sets guaranteed to have well-calibrated prediction probability above a certain threshold 1-ε, with a high probability. This resembles the guarantee presented by van der Laan and Alaa (2024) but for classification. In summary, while the methods have much in common, they originate from regression and classification, respectively, which leads to substantial methodological differences.

Mondrian conformal predictors are category-conditional in the sense that they provide localized validity within a subcategory of data and are thus conditional on features in x Vovk et al. (2022). This framework is similar to some extent to our framework in that it provides its guarantees conditioned on certain subsets of data. However, the conventional way of applying Mondrian conformal predictors is to pre-specify these features. This distinction is important because our framework inherits the property of being well-calibrated conditional on the Venn-Abers taxonomy which, if the model is well-specified, is based on the most influential features for the current prediction. Cross Venn-Abers predictors have been extended to multiclass classification, yielding calibrated probability distributions across multiple classes. While this method extends Venn-Abers to multiclass problems by decomposing them into binary subproblems, each binary predictor focuses only on a pair of labels, and their outputs are aggregated to yield calibrated probability distributions across all classes Manokhin (2017). Majlatow et al. (2025) investigated a variety of calibration techniques, including Venn-Abers predictors, before applying conformal prediction to ensure that prediction probabilities reflect true likelihoods. However, their method only considers binary prediction problems, and because calibration occurs prior to conformal prediction, it does not assign probabilities to prediction sets.

5. Results

The following section presents empirical results for the proposed method. First, we show calibration plots on the simulated data that supports the claim that the method outputs well-calibrated prediction probabilities. Secondly, we evaluate the respective strategies on the MNIST and the CIFAR-10 datasets with respect to calibration metrics, model complexity, calibration plots and examples of images with associated prediction sets and well-calibrated probabilities. Finally, we assess the applicability and implications of the method to AI-driven decision support in a high-stakes domain using actionable prediction sets from the Fitzpatrick17k dataset.

5.1. Results on simulated data

Figure 2 is a calibration plot, sometimes referred to as a reliability diagram. The dotted line in Figure 2 represents perfect calibration, where the predicted probability of $y = 1$ is equal to the true probability of $y = 1$ (see Equation 8). In this framework, $y = 1$ means that the label is contained in the prediction set (see Equation 12). Thus, the y-axis represents empirical coverage across rolling batches of 400 data points. The clusters around 0.62 are a consequence of the simulation scheme. Singleton predictions naturally exhibit this empirical frequency of correct predictions because of the probability of $\frac{37}{60}$ in the simulation scheme. When the tolerance for error becomes less than 60%, the empty sets are replaced with sets of size 2, creating a concentration of predicted probabilities around 0.95 ($\frac{37}{60} + \frac{1}{3}$) rather

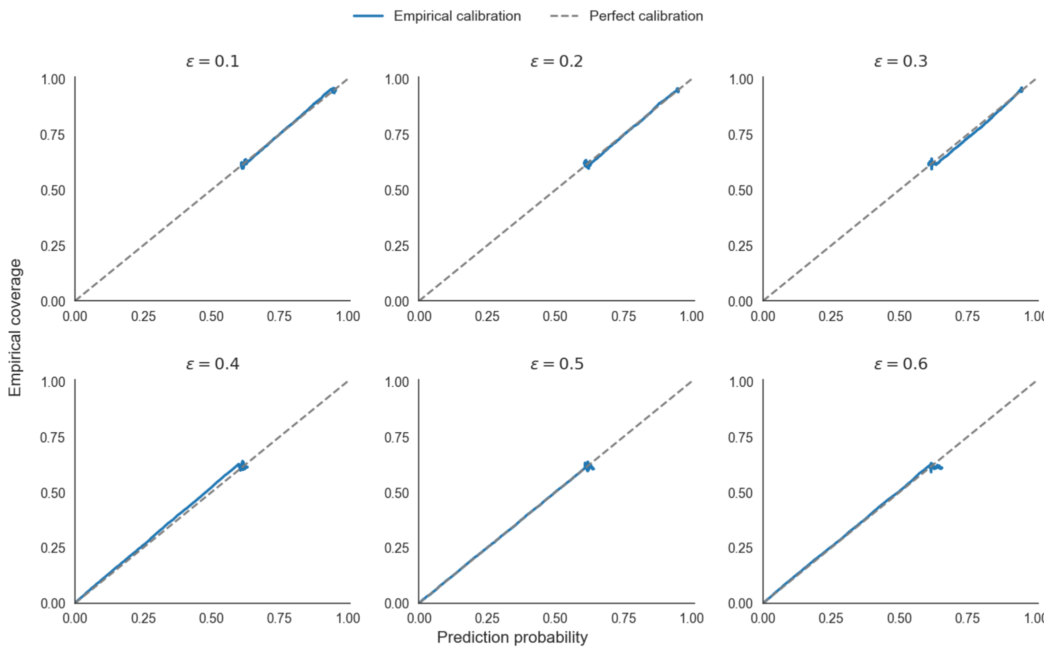

Figure 2: Average prediction probabilities and empirical coverage over 400-sample batches (step size 1) on simulated data for 6 different levels of ε.

than around 0. In Section 3.1, we asserted that the presented framework generates well-calibrated prediction probabilities for large datasets. This means that, for any ε and any predicted probability value, the calibration curve should remain close to the line of perfect calibration. Figure 2 demonstrates that this is indeed the case, supporting the assertion. Notably, some deviation remains, even for such a large dataset.

5.2. Results on the MNIST and CIFAR dataset

Table 1 shows the performance and properties of the ICPs and the IVAPs across the various training strategies and splits with $\varepsilon = 0.01$ for MNIST and $\varepsilon = 0.2$ for CIFAR. The table also shows a comparison on the calibration metrics between the log loss scoring rule (see Equation 9), and the self-consistent scoring rule (see Equation 21). Results for additional ε values on MNIST are provided in Appendix A.1. The IVAPs are evaluated with a weighted average of three metrics: the Brier Score, which measures the mean squared error between the prediction probabilities and actual outcomes Brier (1950); the Balanced Brier Score, which adjusts for class imbalance by weighing the average Brier score across each class equally Gneiting and Raftery (2007); and the Expected Calibration Error (ECE), which quantifies the discrepancy between prediction probabilities and observed accuracy across probability bins Guo et al. (2017). Note that the number of IVAPs that need to be calibrated, and trained for three of the strategies, is equal to the number of unique sets.

The relative performance of the strategies across the two datasets is similar. Retraining the entire neural network achieves the lowest balanced Brier score, indicating the most

Strategy	Unq sets	Brier (LL)	Brier (SC)	Balanced Brier (LL)	Balanced Brier (SC)	ECE (LL)	ECE (SC)
MNIST							
Full retrain	16.25	0.0099	0.0097	**0.1886**	**0.2017**	0.0142	0.0096
Fine-tune		0.0072	0.0072	0.1888	0.2086	0.0084	0.0049
Layer freeze		0.0093	0.0093	0.2215	0.2385	0.0121	0.0080
Prob Pool		**0.0071**	**0.0070**	0.2517	0.2762	**0.0073**	**0.0037**
CIFAR-10							
Full retrain	70.75	0.1376	0.1378	**0.1549**	**0.1559**	0.1204	0.1163
Fine-tune		**0.1182**	**0.1183**	0.1890	0.1912	0.0427	0.0426
Layer freeze		0.1393	0.1393	0.1988	0.2007	0.0777	0.0761
Prob Pool		0.1275	0.1277	0.2348	0.2379	**0.0146**	**0.0188**

Table 1: Performance of IVAPs across adaptation strategies on MNIST and CIFAR-10. Calibration metrics are reported using both the log loss (LL) and self-consistent (SC) scoring rules. Lower values indicate better calibration.

consistent performance across instances within and outside the set prediction. Fine-tuning the model achieves low Brier scores, implying good calibration on average. A notable result is that while probability pooling achieves low Brier scores and ECE, it has a comparatively poor performance on the balanced Brier score, indicating uneven behavior depending on whether the true label falls inside or outside the prediction set. Conversely, the frozen model shows weaker calibration overall but maintains a balanced performance. We revisit this point in the Discussion section. Across 7 of 8 experiments, ECE is lower with the Self-Consistent scoring rule. The trend is most evident on the MNIST dataset. The Brier score is even across the scoring rules while the balanced Brier score is lower for the minmax scoring rule.

Figure 3 shows the calibration plots of each training strategy on the CIFAR-10 dataset, comparing the average empirical coverage of the conformal sets to the average prediction probabilities assigned to those sets across bins of examples with similar prediction probabilities. Because the MNIST model produced very few prediction probabilities below 0.95, the resulting calibration plot covered only a narrow region. For this reason, it is omitted from the main article and presented in Appendix A.

The prediction probabilities are close to the line of perfect calibration. The 1% and 10% mark indicates that the probability pooling strategy assigns less low probabilities than the other strategies, i.e. deviates less often from the coverage probability, 80%. The same trend appears for MNIST, as illustrated in Figure 5 of Appendix A1. Note that Probability pooling has less low probability predictions even though the line extends longer than the others. This is merely a consequence of there being a bin with more than 550 data points and satisfying the 0.02 difference threshold.

Figure 4 shows examples of images from the MNIST dataset with prediction sets and their associated IVAP prediction probabilities. The examples are grouped into three different categories, ranging from low to high IVAP prediction probabilities.

We discuss these examples further in the Discussion section.

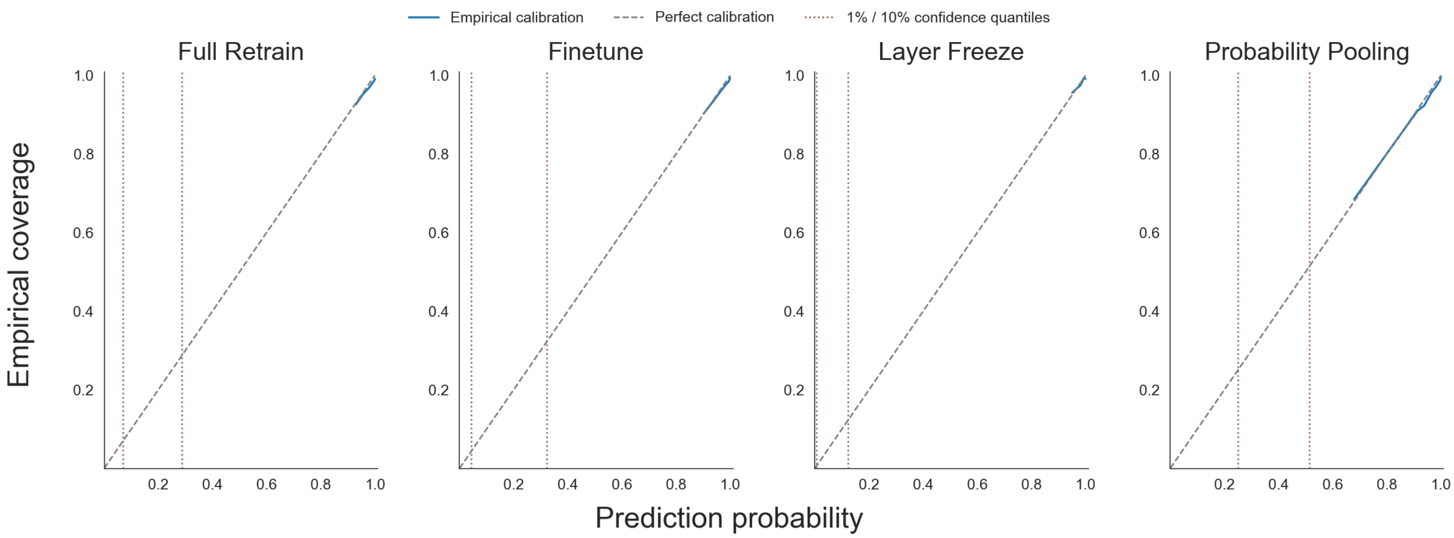

Figure 3: Average prediction probabilities and empirical coverage over batches of 550-1000 datapoints from CIFAR. Bins where the difference between the largest and smallest value was larger than 0.02 were removed. Markers highlight where cumulative proportions exceed 1% and 10%.

Training strategy	ε	**#Unique sets**	**Brier score**	**Balanced Brier score**	**ECE**
Probability Pooling	0.1	2629	0.0834	0.4010	0.0162
Probability Pooling	0.2	2037	0.1563	0.3035	0.0294

Table 2: Performance on the clean Fitzpatrick17k dataset

5.3. Results on the real-world clean Fitzpatrick17k dataset

Table 2 presents the results of the probability-pooling strategy, for two choices of ε, applied to the softmax scores from Cortes-Gomez et al. (2025b), which were computed on the clean Fitzpatrick17k dataset.

As indicated by Table 2, the number of unique sets is substantially larger for the clean Fitzpatrick17k dataset compared to the MNIST and CIFAR-10 datasets. This is expected since there are 48 labels, and thus 2^{47} possible prediction sets in the clean Fitzpatrick17k dataset. For $\varepsilon = 0.2$, the Brier score and the ECE are worse compared to $\varepsilon = 0.1$, while the balanced Brier score is better.

The actionability of a prediction depends on the context. The risks involved should affect the level of confidence a healthcare professional desires from a model. The labels in the clean Fitzpatrick17k dataset ranges from very serious to mild conditions. Thus, the prediction probability required for an action likely will differ between labels. For the sake of simplicity we treat all samples with the same IVAP probability threshold. Table 3 gives an overview of the proportion of the actionable prediction sets for different thresholds.

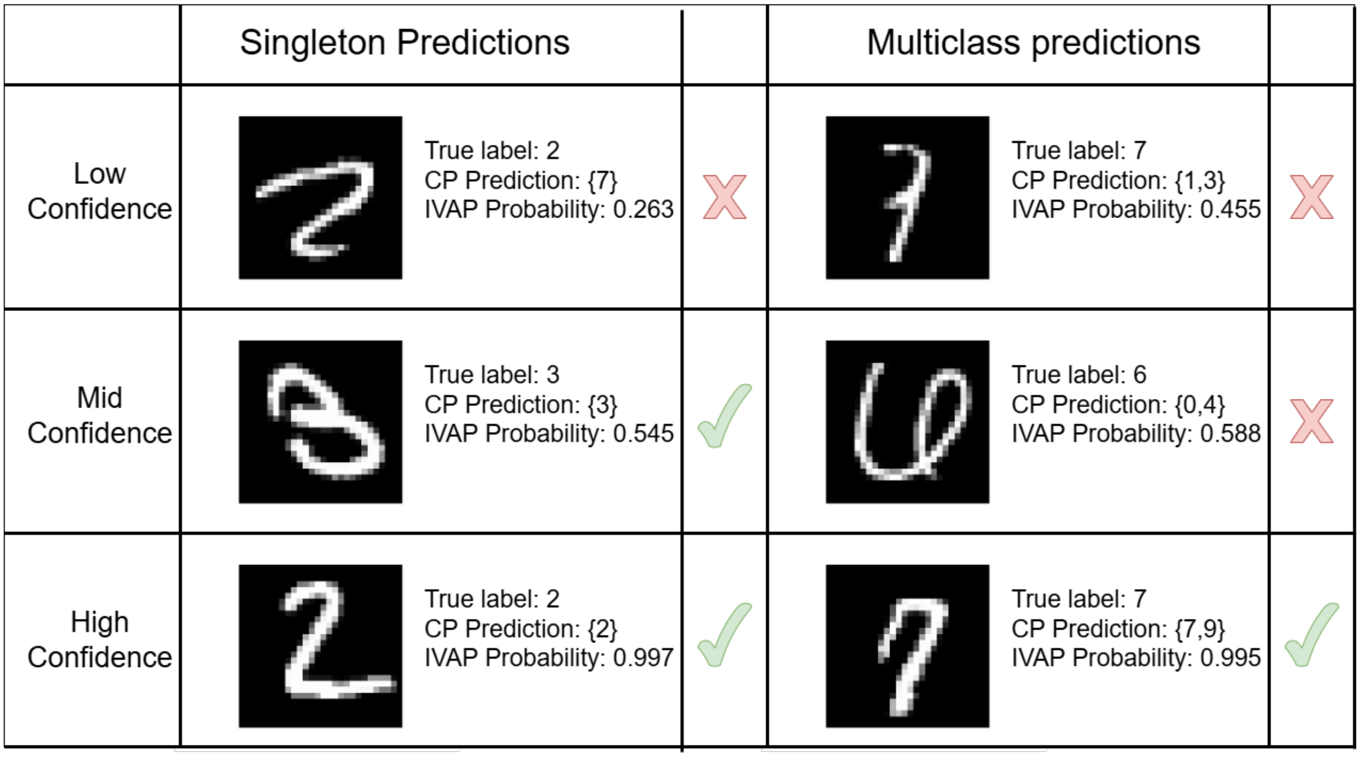

Figure 4: Examples of well-calibrated prediction probabilities from an IVAP applied to prediction sets from a conformal predictor trained on the MNIST dataset.

Probability Threshold	$\boldsymbol{\varepsilon = 0.1}$	$\boldsymbol{\varepsilon = 0.2}$
0.70	0.0910	0.2077
0.80	0.0807	0.1570
0.90	0.0487	0.0667
0.95	0.0087	0.0090
0.99	0.0000	0.0000

Table 3: Proportion of data with actionable superclasses across different IVAP probability thresholds for two epsilon levels.

It is noteworthy that there are more actionable sets for the higher ε even when the threshold is equal to, or above, $1 - \varepsilon$.

6. Discussion

Trustworthy AI models are indispensable in high-stakes domains where errors carry significant consequences. We have demonstrated that it is possible to obtain reliable prediction probabilities for set predictions and that these probabilities can provide direct decision support in real-world high-stakes domains.

Table 1 points towards a trade-off relationship between model complexity and performance on the balanced Brier score for the four strategies. Notably, pooling the probabilities from the conformal predictor—requiring no retraining of the IVAPs—achieves strong performance in terms of Brier score and expected calibration error (ECE). Although it attains the highest balanced Brier score, its competitiveness, even on this metric, is striking. The observed discrepancy between performance on balanced and unbalanced metrics can be, at least partially, attributed to the fact that the probability pooling strategy relies on the heuristic probabilities of the base model. Broadly, this coupling likely increases the agreement between the ICP and the IVAPs compared to the other strategies, which in turn tends to produce higher predicted probabilities. In our setting—where probabilities are typically above 0.5—this naturally leads to less balanced predictions. This dependence also induces a lower bound on the inputs to the IVAP (see equation 22), which is one concrete way in which it can influence the prediction probabilities toward higher values. This trend is evident in Figure 3, where the 1 and 10% marks are significantly higher for the probability pooling strategy compared to the other training strategies. For $\varepsilon = 0.2$, approximately 80% of samples will be ones, and when the majority of the samples are ones, any deviation from a high prediction probability will be costly for the unbalanced metrics unless it is executed with high precision. Notably, the full retraining strategy displays the exact opposite trend, having the best performance on balanced Brier score but showing moderate results on the unbalanced metrics.

These findings might explain the probability pooling strategy's strong performance on the unbalanced metrics. However, the most desired behavior of a probabilistic predictor is that it can deviate from the empirical frequency when warranted. The balanced Brier score demonstrates that this is a key limitation of the probability pooling strategy, whose performance is, to a larger extent than the other strategies, driven by predicting high probabilities for the dominant class. To illustrate why this is undesirable, we consider a hypothetical model that outputs random heuristic probabilities. Within a Venn–Abers machine, such a model would still yield well-calibrated prediction probabilities. However, it would achieve this by assigning nearly uniform probabilities across all prediction sets, effectively mirroring the empirical frequency of ones in the calibration set, which in this framework will be close to the coverage probability $1 - \varepsilon$. These prediction probabilities, while theoretically being well-calibrated, would in practice add nothing to the marginal coverage guarantee of the conformal predictor. Although this is not the case for the probability pooling strategy, which performs competitively even on the balanced Brier score, it illustrates why a decrease in deviation from empirical frequency is undesirable. It is also worth noting that this generally yields better performance on unbalanced metrics when the label distribution is heavily skewed. This is supported by the results in Table 2, where the Brier score improves for lower ε and the balanced Brier score improves for higher ε. The trend can also be observed by comparing performance across different values of ε in table 4 in Appendix A.1.

Another effect of the lower bound in 22 seems to be that the probability pooling strategy is less likely to output prediction probabilities where $p < \frac{|\Gamma|}{k}$. For example, for a singleton prediction on the MNIST dataset it is impossible for the prediction probability of the probability pooling strategy to be less than 0.1 before calibration, and is thus probably also unlikely after calibration. The other strategies have no such lower bound and it is most likely therefore we observe more of these inconsistent outputs. This might be seen as a strength of the probability pooling strategy.

The practical utility of the proposed framework when applied to the clean Fitzpatrick17k skin tone dataset can be observed in Table 3. Despite the superclasses containing at most three labels and the total label space including 48 classes, a substantial minority of data points can be assigned actionable recommendations. It is plausible that, for many problems with smaller label spaces, the proportion of actionable prediction sets would be significantly higher. It is notable that there are fewer actionable sets for $\varepsilon = 0.1$ even when the threshold is equal to, or above: $1 - \varepsilon$. This occurs because some prediction sets are subsets of an actionable superclass for a lower ε. When ε is higher, a few of these set predictions contain additional labels within the same superclass, allowing assignment of a higher prediction probability to the same action. This result shows the need for more specific construction of prediction sets tailored to the specifics of each example and motivates the extension of the presented framework described in the future work section.

Crucially, the guarantees from both ICP and IVAP, and thus the inherited guarantees in this framework, depend on the size of the calibration set. Equations 6 and 5 describe the relationship between the size of the calibration set and the probability that the marginal coverage guarantee will hold. Equation 7 shows the lower bound on the necessary size of the calibration set to achieve at least a probability θ for the guarantee to hold. Similarly, p_0 and p_1 are only close, and thus both well-calibrated, for large datasets as described in the introduction to IVAPs. In big data settings, this is typically not a problem.

The full implementation of the framework can be computationally intensive. In the worst case scenario, $min(2^{k-1}, n_{test})$ models are required. For problems with large k and/or large n this might be impractical when the models require long training times. The training strategies presented in this work address this issue and mitigate the computational burden to varying degrees. The layer-freeze strategy reduces the number of trainable parameters by multiple orders of magnitude. While being reasonably competitive for the problems presented in this article, it is possible that the layer-freeze strategy will perform much worse for more complex problems, whereas fine-tuning may remain competitive with full retraining. However, conceptually, it is not a big leap for a model that can differentiate the label space effectively into k labels to group these labels into two categories. This means that the layer-freeze strategy, and indeed the probability pooling strategy, might remain competitive.

The probability pooling strategy eliminates the need for model retraining entirely. The only computation required is calibrating twice as many isotonic regressions as there are unique prediction sets, which is computationally inexpensive and tractable even for hundreds of thousands of sets. However, as discussed above, the strategy has the undesired property that it deviates less from the empirical frequency of the majority class. Despite this limitation, the strategy remains competitive and potentially invaluable in settings involving complex problems and large label spaces. Nonetheless, this weakness warrants careful consideration when calibration precision is critical.

The optimal choice of strategy is context-dependent and influenced by factors such as size and complexity of the dataset, model architecture, available computational resources, decision support requirements, and the required differentiation in the prediction probabilities. More experiments in various contexts are needed to make any statements about what strategies are apt for what context. Still, we provide some carefully considered recommendations based on the available results. If computational resources are not a constraint, we recommend fully retraining the IVAP models. If resources are limited but available and the problem is complex, fine-tuning the base ICP model is likely preferred. For problems with very large label spaces and high complexity, pooling probabilities from the base ICP model may be the only practically feasible option.

We end this section with a brief discussion of the examples in Figure 4. In the high confidence singleton prediction, the prediction probability reflects that the image is clearly showing a 2. The high confidence multiclass prediction reflects that the image shows a 7 or a 9 almost certainly, but which one is not clear. The mid confidence singleton prediction is correct, but one can see why the prediction probability is lower, the image could have been labeled a 9 or an 8. The remaining prediction sets do not contain the true label, but one can still understand why the model selected the labels it did. At the same time, the IVAP models appropriately signal lower confidence in these predictions.

7. Future work

For any given data point, it is possible to utilize the presented framework to find the smallest possible set whose prediction probability exceeds a threshold, $1 - \varepsilon$. This enables an extension of the method to assign prediction sets with a high prediction-conditional probability. However, such an approach will introduce multiple-testing problems which need to be mediated. In the near future, we aim to pursue this line of research.

The framework can be extended to any algorithm whose output space can be interpreted as a probability distribution. Consequently, it is applicable even to probabilistic regression models. However, since there are an infinite number of possible partitions of regression label spaces, calibration and any necessary training must occur contemporaneously with prediction. This suggests that probability pooling is the most practical strategy unless models are sufficiently small. For probability pooling, the probability density function of the regression model would be integrated within the prediction band to form heuristic probability estimates for the label being in the set.

The layer-freeze and fine-tune strategies require little code modification in deep learning libraries, such as PyTorch, TensorFlow, or Keras, since the frameworks are highly flexible. In contrast, libraries such as Scikit-learn, where model configurations are less adaptable, may require feature extraction to reduce training costs. Any kind of feature extraction could be employed in a similar fashion as presented for the layer-freeze strategy in the Method section. For example, if the base ICP model is a random forest, one could extract all the leaves from it and use them to train a logistic regression model. Note that both probability pooling and full retraining strategies are compatible with any model that provides class probability estimates, regardless of library or architecture.

One promising application of the method is in the treatment of antibiotic-resistant bacteria, a global challenge that continues to grow in complexity and urgency. Rapidly identifying an effective antibiotic is crucial, as delays can directly worsen patient outcomes. While sequencing the DNA of bacterial cultures can support this process, culturing itself often requires several days Cleveland Clinic (n.d.). Furthermore, once bacteria are identified, resistance profiles can vary substantially within the same species Sun et al. (2022). Indiscriminate antibiotic use is discouraged because it accelerates resistance Organization (2025). In this context, tools that quantify uncertainty in predicted resistance profiles could benefit from the conceptualization of actionable superclasses. Even when the exact resistance profile is uncertain, there may still be high confidence that the true profile lies within a subset of profiles that share the same viable treatment options.

In classification settings with large label spaces, conformal prediction provides sets of plausible labels, allowing domain experts to further refine the classification based on contextual knowledge. Our framework extends this capability by associating conditionally calibrated probabilities with each set. For example, high-probability sets can be considered informative and prioritized for expert review, while low-probability sets can be rejected. Alternatively, flagging uncertain instances may indicate the need for experts to look outside of the prediction set for the correct label, drawing attention to ambiguous or unusual samples.

The use cases discussed above represent only a subset of the potential application domains and methodological extensions of the framework. Future research should explore additional domains, problem settings, and algorithmic developments to broaden the applicability of the approach. Finally, there may exist strategies beyond those presented in this paper to further reduce computational burden. Identifying and evaluating such strategies remains an important direction for future work.

8. Conclusion

Uncertainty quantification is necessary in high-stakes domains. We have presented a framework that generates well-calibrated probabilities for set predictions and thus specifies the confidence in each individual set prediction. Further, we have demonstrated how these probabilities can be used for decision support. The framework inherits the guarantees from both conformal prediction and the Venn-Abers framework. We have designed and evaluated various strategies to mitigate the computational burden of the framework. The strategies range from complete retraining of a new model for each unique prediction set, to fine-tuning of varying degrees, to simply calibrating the aggregated heuristic prediction probabilities of the base ICP model. The best choice of strategy depends on the context, but it is noteworthy that the probability pooling strategy is feasible in almost any setting. We advocate that the framework is relevant in any high-stakes domain where machine learning is deployed for classification.

9. Code and Data availability

The associated code and data used in the study are available at https://github.com/OskarGustafsson123/CP-VA and https://zenodo.org/records/17937611 (DOI: 10.5281/zenodo.19681063).

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  - Styrbjörn Käll
- **Formal Analysis** — Applying statistical or computational analyses.
  - Oskar Gustafsson
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  - Johan Hallberg Szabadváry
  - Styrbjörn Käll
- **Methodology** — Designing methods or analytical approaches.
  - Oskar Gustafsson & John Pavia
  - Stybjörn Käll
- **Software** — Developing or maintaining code and tools.
  - Oskar Gustafsson
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Strategy	Epsilon	Unique Sets	Brier Score	Balanced Brier Score	ECE
Full Retrain	0.05	24	0.0063	0.1963	0.2605
Fine-tune			0.0056	0.2602	0.2317
Layer Freeze			0.0083	0.3399	0.3013
Probability Pooling			0.0063	0.3940	0.1436
Full Retrain	0.15	67	0.0107	0.1208	0.1641
Fine-tune			0.0086	0.1425	0.1093
Layer Freeze			0.0150	0.2076	0.2205
Probability Pooling			0.0117	0.2413	0.1121

Table 4: Comparison of strategies across different epsilon values and unique sets, with Brier Score, Balanced Brier Score, and Expected Calibration Error (ECE).

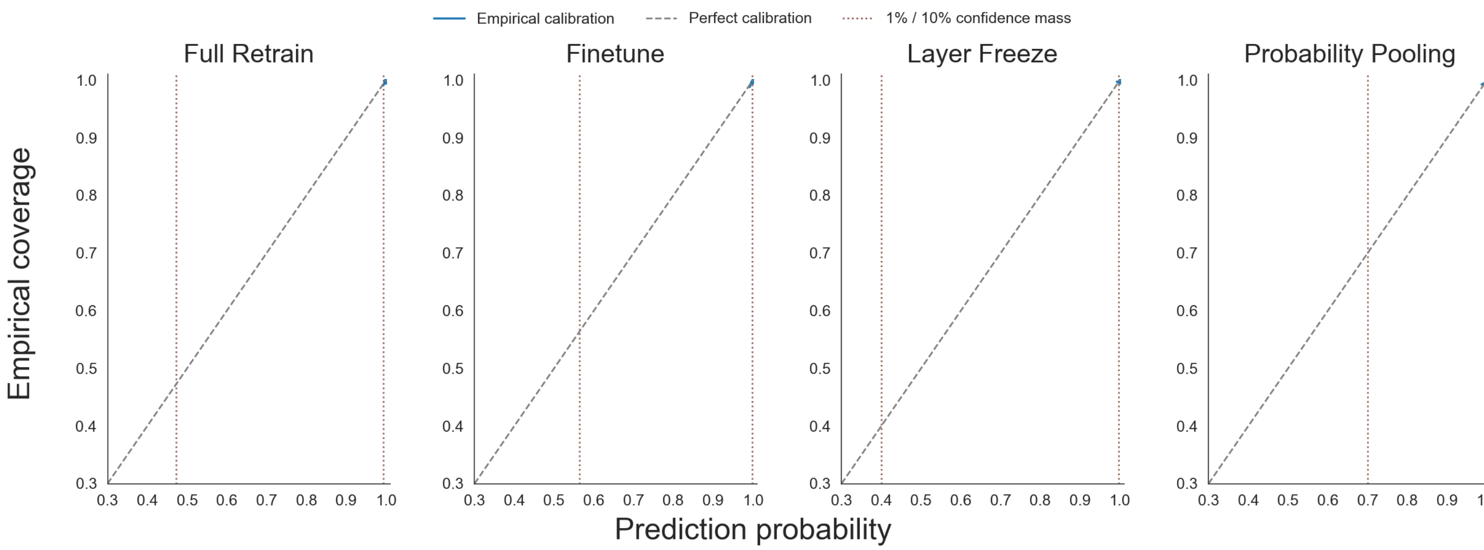

Figure 5: Calibration curves for MNIST across the four different strategies.

Appendix

A.1 MNIST Additional Results

A2. Algorithm

Here, the method is described on an algorithmic form along with a description that is meant to provide intuition for the various steps in the algorithm. In the full implementation, Algorithm 2 is executed only once while Algorithm 3, depending on the chosen strategy, is run once for each datapoint. Note that the algorithm differs from the formal description in that only the IVAPs corresponding to the predictions sets outputted by the ICP on the test set are being trained. This modification saves time and is the most likely way the method will be applied in experimental settings.

Algortithm 1

Algorithm 1 Create Binary Labels from Prediction Sets

Require: $Y = \{y_i\}_{i=1}^{n}$, list of unique prediction sets $\mathcal{G}^{\varepsilon} = \{\Gamma^{\varepsilon}(x_i)\}_{i=n+1}^{l}$
Ensure: A binary label matrix Y_{new} of shape $n \times 2^{k-1}$

for $j = 1$ to l **do**
  if $\Gamma^{\varepsilon}(x_j) \neq \emptyset$ **then**
    for $i = 1$ to n **do**
      if $y_i \in \Gamma^{\varepsilon}(x_j)$ **then**
        $Y_{\text{new}}[i][j] \leftarrow 1$
      else
        $Y_{\text{new}}[i][j] \leftarrow 0$
      end if
    end for
  end if
end for

In Algorithm 1, corresponding to equation 12, the labels for the binary prediction problems are created. This is achieved by going through all unique prediction sets and all labels and creating a matrix where the ith, jth element is a 1 if the $i : th$ label is inside of the $j : th$ set. If $\Gamma^{\varepsilon}(x_j)$ is empty, it means that this set was never suggested by the ICP, or that it was suggested but is the empty set. Either way, we can ignore it.

Algorithm 2

The output of algorithm 2 is a matrix called S and a list called M. In S, each column corresponds to a datapoint, and the row corresponds to a set. Each element is a model's heuristic prediction probability that the respective datapoint is inside the respective set. M is a list of models and the index of each model corresponds to the index of each set from $\mathcal{G}^{\varepsilon}$, a list with all unique sets from the ICP. The function, index_{Π_A}, applied in the algorithms refer to equation 11 and is used repeatedly to sort the models and calibration scores based on the prediction set they belong to. As mentioned above, inductive conformal prediction is being run initially to limit the number of IVAPs being trained. One model is trained for each set, and the calibration probabilities that will be used in the isotonic regression are calculated.

Algortithm 3

Algorithm 3 starts with applying an ICP to a new datapoint. The model, the calibration scores for the relevant set, and X_{new} are used in an IVAP to assign conditionally calibrated probabilities to the corresponding prediction set. If a_1 is in $\Gamma^{\varepsilon}(x_{new})$, the model has been trained on whether the label is inside $\Gamma^{\varepsilon}(x_{new})^C$ or not, and $p0$ and $p1$ need to be adjusted.

Algorithm 2 Preparation of models and heuristic prediction probabilities

Require: Training data $D_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{m}$, CP calibration data $D_{\text{cal}}^{\text{CP}} = \{(x_i, y_i)\}_{i=m+1}^{n_1}$, VA calibration data $D_{\text{cal}}^{\text{VA}} = \{(x_i, y_i)\}_{i=n_1+1}^{n}$, test data $X_{\text{test}} = \{x_i\}_{i=n+1}^{t}$, model f, miscoverage level $\varepsilon \in (0, 1)$

Ensure: Matrix S, list of models M

```
Apply ICP on X_test using calibration set D_cal^CP to obtain prediction sets Γ^ε(X_test) = {Γ^ε(x_i)}_{i=n+1}^t
𝒢^ε ← array of size 2^{k−1} initialized with empty values
for j = 1 to |Γ^ε(X_test)| do
    Γ^ε(x_j) ← prediction set for test point x_j
    index ← index_{Π_A}(Γ^ε(x_j))
    if a_1 ∉ Γ^ε(x_j) then
        𝒢^ε[index] ← Γ^ε(x_j)
    else
        𝒢^ε[index] ← Γ^ε(x_j)^C
    end if
end for
Y_train,new ← Algorithm1({y_i}_{i=1}^m, 𝒢^ε)
Y_cal,new ← Algorithm1({y_i}_{i=n_1+1}^n, 𝒢^ε)
M ← array of size 2^{k−1} initialized with empty models
S ← array of size (2^{k−1}, |D_cal^VA|) initialized with empty values
for j = 1 to |𝒢^ε| do
    if 𝒢^ε[j] ≠ ∅ then
        Γ^ε(x_j) ← 𝒢^ε[j]
        index ← index_{Π_A}(Γ^ε(x_j))
        Train model m on X ∈ D_train and Y_train,new[:, index]
        M[index] ← m
        S[index] ← m(X ∈ D_cal^VA)
    end if
end for
return S, M
```

Algorithm 3 Assigning conditionally calibrated probabilities to prediction set

Require: New input x_{new}, unique prediction sets $\mathcal{G}^{\varepsilon}$, VA calibration probabilities S, list of models M, VA calibration labels $Y_{\text{cal,new}}$, miscoverage level $\varepsilon \in (0,1)$

Ensure: Prediction set $\Gamma^{\varepsilon}(x_{\text{new}})$, adjusted prediction probabilities $p_0^{\text{Adjusted}}, p_1^{\text{Adjusted}}$

$\Gamma^{\varepsilon}(x_{\text{new}}) \leftarrow \text{ICP}(x_{\text{new}}, \varepsilon; D_{\text{cal}}^{\text{CP}})$
index $\leftarrow \text{index}_{\Pi_A}(\Gamma^{\varepsilon}(x_{\text{new}}))$
$\text{s}_{x_{\text{new}}} \leftarrow M[\text{index}](x_{\text{new}})$
$(p_0, p_1) \leftarrow \text{IVAP}(\text{S}[\text{index}], \text{s}_{x_{\text{new}}}, Y_{\text{cal,new}}[:, \text{index}])$
if $a_1 \notin \Gamma^{\varepsilon}(x_{\text{new}})$ **then**
    $p_0^{\text{Adjusted}} \leftarrow p_0$
    $p_1^{\text{Adjusted}} \leftarrow p_1$
else
    $p_0^{\text{Adjusted}} \leftarrow 1 - p_1$
    $p_1^{\text{Adjusted}} \leftarrow 1 - p_0$
end if
return $\Gamma^{\varepsilon}(x_{\text{new}}), p_0^{\text{Adjusted}}, p_1^{\text{Adjusted}}$